



COLLEGE OF
COMPUTER STUDIES

Week 5

FOPR111 – Fundamentals of Programming

C OPERATORS

Objectives:

By the end of this lesson, you should be able to:

- Familiarize with different operators
- Learn to write programs using arithmetic and assignment operators
- Understand the creation of expressions



C OPERATORS

C Operators

Operator is a symbol that is used to perform mathematical or logical manipulations in the program. C language has the following operators:

Arithmetic Operators

These are used to perform basic mathematical calculations like addition, subtraction, multiplication, division and modulus operations in an expression.

Operator	Description	Example
+	Addition or unary plus	$C=A+B$ gives sum of $A+B$
-	Subtraction or unary minus	$C=A-B$ gives difference of $A-B$
*	Unary or multiplication	$C=A*B$ gives multiplication of $A*B$
/	Division operator	$C=A/B$ gives quotient value
%	Modulo Division	$C=A\%B$ gives remainder value

Relational Operator

These operators are used to compare the value of two variables (or) operands.

Operator	Description	Example
==	Equal to	5 ==3 returns false (0)
>	Greater than	5>3 returns true (1)
<	Less than	5<3 returns false (0)
!=	Not equal to	5!=3 returns true(1)
>=	Greater than or equal to	5>=3 returns true (1)
<=	Less than or equal to	5<=3 return false (0)

Logical Operators

These operators are used to compare two or more relational expressions.

Operator	Description	Example
&&	Logical AND	If c=5 and d=2 then, ((c==5) && (d>5)) returns false.
 	Logical OR	If c=5 and d=2 then, ((c==5) (d>5)) returns true.
!	Logical NOT	If c=5 then, !(c==5) returns false.

Bitwise (or) Binary Operators

C has distinction of supporting special operators known as bitwise operators for manipulation of data at bit level. These operators are used for testing the bits and shifting them right to left or left to right.

Operator	Description	Example(A=12, B=2)
&	Binary AND	A&B gives 0
	Binary OR	A B gives 14
^	Binary EX-OR	A^B gives 14
~	Binary Complement	~B gives -3 hint: -(n+1)
<<	Left shift	A<<2 gives 48
>>	Right shift	B>>2 gives 0

NOTE

- Left shifting one time means, it is equal to multiplying the value by 2 one time.
- Right shifting one time means, it is equal to dividing the value by 2 one time.

Assignment Operators

This operator is used to assign the values for the variables in the programs

Operator	Description	Example
=	Simple assignment	A=B; i.e. B value is assigned to A
+=	Addition and assignment	A+=B is same as A=A+B
-=	Subtraction and assignment	A-=B is same as A=A-B
=	Multiplication and assignment	A=B is same as A=A*B
/=	Division and assignment	A/=B is same as A=A/B
%=	Modulus and assignment	A%=B is same as A=A%B
<<=	Left shift and assignment	A<<=B is same as A=A<>=	Right shift and assignment	A>>=B is same as A=A>>B
&=	Binary AND and assignment	A&=B is same as A=A&B
=	Binary OR and assignment	A =B is same as A=A B
^=	Binary EX-OR and assignment	A^=B is same as A=A^B

Increment or Decrement Operators

These operators are used to increment or decrement by 1 in an expression.

Operator	Description	Example
++operand	Unary pre-increment	A=++10; first evaluates 11 and assigns A=11
--operand	Unary pre-decrement	A=--10; first evaluates 9 and assigns A=9
operand++	Unary post-increment	A=10++; first assigns A=10 then increments A by 1
operand--	Unary post-decrement	A=10--;first assigns A=10 then decrements A by 1

Thank you

Let's do something great